m=4, l=6 (m*l = 24): an unsatisfiable instance, and why its degree-4 closure did not finish here

Write-once record, 2026-10-26. Follows `crosscheck-hky-par-20261020.md` (unchanged). Observations are measured; derived numbers are labelled. Raw files and hashes: `raw-20261026/`. Observations only: no claim about the degree of any instance is made.

Why n = 17

The m=4 cells use N = 2n + 4l variables. A degree-4 refutation needs an **unsatisfiable** system, and the exact ordered solution count of a draw is about 2^(m*l - n) (measured counts below). Unsatisfiable draws therefore thin out roughly by a factor of two per unit of m*l - n.

- At (n, l) = (13, 5), surplus 7: one unsatisfiable draw in 19 (the earlier record, draw 18).
- (n, l) = (13, 6), surplus 11: a first attempt (`hky_par newcells --cells 4:13:6`) ran 31 CPU-minutes in the single-threaded draw phase with no result and was stopped. Not a measurement of anything but cost.
- Measured exact (ordered) solution counts, seed 20261016: (16, 6): 276, 240, 216 at 92 s per count; (15, 6): 288, 1290, 180 at 47 s per count. Unsatisfiable draws there are not findable (derived: about 1 in 20,000 at n = 16).

Search method (scratch tool, not pushed to crypto)

`hky_sys findunsat` (diff in `raw-20261026/hky_sys-vs-hky_par.diff`, 119 lines against `examples/hky_par.rs` at crypto 981f80710b7e) replays the `newcells` draw stream (`cell_seed`, `ladder_sample`) and decides each draw with an **early-exit** version of `chained_solution_count`'s search, which returns at the first solution. Draws are split over workers by `draw % workers`; each worker replays the same generator stream.

- Validation against the exact counter: at (13, 5) draws 0-18 all agree, and draw 18 is the unsatisfiable one found by the earlier run. At (16, 6) draws 0-2 and (17, 6) draw 0 the early-exit result is "has solution" with exact counts 276 / 240 / 216 / 48. Early-exit cost: median < 0.05 s at (13, 5); 1.5-4.1 s at (16, 6); 5 s at (17, 6) versus 92-190 s for the exact count.
- Four workers, `--cell 4:17:6 --seed 20261016`: worker 1 found **draw 57** with no solution, after 198 s, and the exact counter then returned **0 solutions** for it. Draw 57 is not shown to be the lowest-index unsatisfiable draw (the other workers were stopped at their own draws).

The instance (`raw-20261026/unsat_m4_n17_l6.jsonl`)

m = 4, n = 17, l = 6, seed 20261016, draw 57, v_basis = [82767, 93651, 44058, 109552, 7169, 121640], x_R = 56927, exact solution count 0. System built by `hky_par`'s own `system()` (b fixed to 1): 58 variables.

Closure measurements (`hky_par ladder ... --m 4`, 4 threads, same code as the earlier record)

degree	result	dim	columns
2	not refuted	74	1,712
3	not refuted	3,299	32,568
4	**did not complete**: allocation failed at the 14.5 GB address-space cap (`prlimit --as`) after 1,973 s; last sampled resident memory 11.65 GB	n/a	456,838

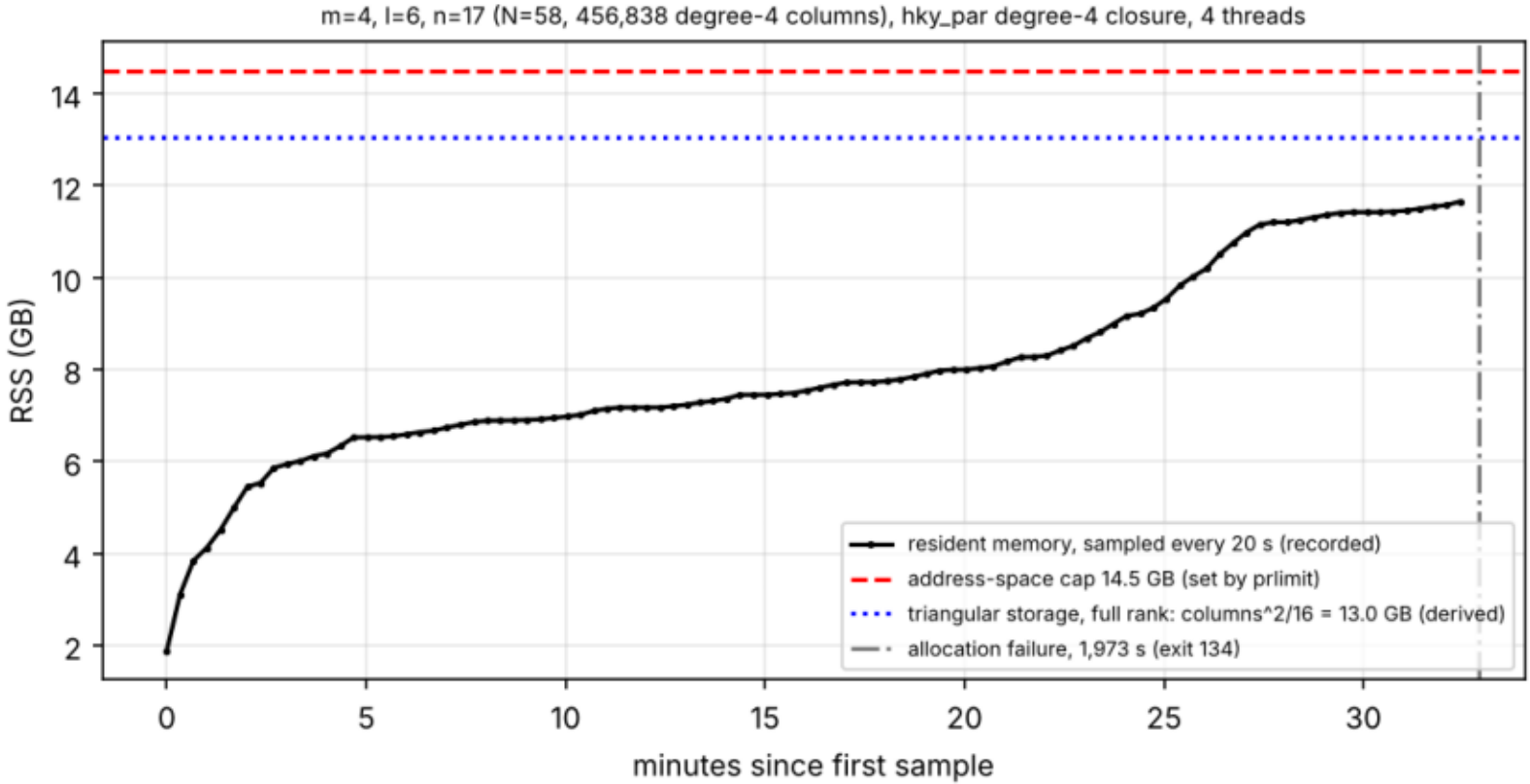
The degree-4 outcome is "not obtained", not "refuted" and not "not refuted". The RSS samples (every 20 s, 98 samples) are in `c4.rss` and plotted in `m4l6-rss.svg`. The cap limits virtual address space, so the sampled resident peak (11.65 GB) is a lower bound for the memory the closure would have asked for.

Derived: what the closure needs (extrapolation; 2.5x more columns than the largest completed cell)

- Columns for N = 58 at degree 4: 456,838 (exact). Row-triangular storage at full rank: columns^2 / 16 bytes = 13.0 GB. The one completed degree-4 m=4 run (N = 46, 179,447 columns) peaked at 3.26 GB = 1.62 x its triangular figure (2.0 GB). Applying that factor gives about 21 GB; the factor at this size is unknown.
- Time: 613 s at N = 46 on 4 threads. Scaling by (columns ratio)^2 to ^3 gives 66 min to 2.8 h. Not validated.
- Planning figure for a host: 32 GB RAM, 4+ cores. An unvalidated choice, not a bound.
- Why no smaller cell works: memory limits N to about 54 (n = 15) here, where the unsatisfiable-draw rate is about e^-21 per draw by the solution-count estimate; n = 17 gives a findable rate but N = 58. This is the two-sided constraint on an m = 4, l = 6 cell in this container.

Limits

One instance, one run, no variance. No degree-4 result at m*l = 24. The early-exit decision procedure was validated on 19 + 4 draws, not exhaustively; the instance itself is certified by the **exact** counter (0 solutions). The `drawcount`/`--check` raw JSON lines were not saved; `raw-20261026/drawcount-and-check.txt` has the parsed values.


m=4, l=6, n=17 (N=58, 456,838 degree-4 columns), hky_par degree-4 closure, 4 threads
RSS (GB)
14
12
10
8
6
4
2
0
5
10
15
20
25
30
minutes since first sample
resident memory, sampled every 20 s (recorded)
address-space cap 14.5 GB (set by prlimit)
triangular storage, full rank: columns^2/16 = 13.0 GB (derived)
allocation failure, 1,973 s (exit 134)